

The Block Universe Confronted by TNA

Eternalism, Closure, and the Ontology of Realizability

Abstract

The Block Universe interpretation of relativity proposes that spacetime is a fully realized four-dimensional manifold in which past, present, and future coexist equally. Under this ontology, temporal becoming is treated as either subjective illusion or epistemic limitation. While mathematically elegant and deeply compatible with relativistic geometry, eternalism faces persistent difficulties concerning temporal asymmetry, conscious experience, novelty, and the realizability of trajectories.

This paper analyzes eternalism through the framework of the Theory of Axiomatic Necessity (TNA). TNA distinguishes between local operational dynamics (N_0) and non-derivable realizability structure (N_1). From this perspective, the Block Universe is reinterpreted not as a fundamentally false geometry, but as a macroscopic approximation emerging from deeper Planck-scale realizability selection processes.

A new ontological formulation — Model 2 — is introduced. Model 2 proposes that spacetime is not a completely pre-existing object but a continuously reselected structure whose realizability depends on discrete non-derivable selection events occurring at Planck-scale intervals. Relativistic geometry remains locally valid between selection events, while global closure becomes structurally impossible.

Under this interpretation, becoming regains ontological status, temporal asymmetry becomes structural rather than purely thermodynamic, and consciousness emerges as a resolution-limited interface to realizability dynamics rather than a passive observer embedded in a static manifold.

1. Introduction

The emergence of relativistic spacetime radically transformed modern ontology. Following Einstein and Minkowski, time increasingly came to be interpreted not as a flowing process but as a geometric dimension integrated into a four-dimensional manifold. This interpretation naturally gave rise to eternalism, or the Block Universe view, according to which all events —

past, present, and future — possess equal ontological reality.

Within this framework, temporal becoming is not fundamental. The “present” becomes observer-dependent, while the flow of time is reduced to either psychological illusion or informational limitation.

This view has been defended in multiple forms by philosophers such as:

- J.M.E. McTaggart,
- D.H. Mellor,
- Theodore Sider,
- and other proponents of B-theory temporal ontology.

At the same time, critics including:

- A.N. Prior,
 - David Zimmerman,
 - Henri Bergson,
 - Alfred North Whitehead,
 - and more recently Craig Callender,
- have argued that eternalism struggles to adequately explain the phenomenology of becoming, temporal asymmetry, and conscious experience.

The central question remains unresolved:

How can a completely static spacetime manifold generate the persistent appearance of becoming, novelty, and irreversible experience?

This paper proposes that the problem is not primarily physical but structural.

Using the framework of the Theory of Axiomatic Necessity (TNA), eternalism is reinterpreted as a valid macroscopic geometric approximation that becomes ontologically misleading when treated as fundamentally complete.

TNA argues that local physical dynamics alone cannot fully generate realizability. Every sufficiently complex system requires non-derivable closure conditions — termed N_1 structures — that stabilize trajectories and constrain multiplicity.

Under this interpretation, spacetime is not fundamentally a completed object. Rather, the apparent Block Universe emerges from continuous Planck-scale realizability selection processes.

2. Eternalism and the Classical Block Universe

The Block Universe interpretation arises naturally from relativistic spacetime geometry.

In Minkowskian form, spacetime is represented as:

$$M_4 = (x, y, z, t)$$

where temporal coordinates possess the same ontological status as spatial coordinates.

Under eternalism:

- all events coexist equally,
- temporal passage is not fundamental,
- and the distinction between past, present, and future becomes perspectival rather than ontological.

This interpretation aligns naturally with:

- Einstein's relativity,
- four-dimensionalism,
- and B-theory temporal ontology.

The Block Universe possesses several major strengths:

- relativistic invariance,
- elegant causal structure,
- geometric determinism,
- and removal of privileged reference frames.

However, it also generates major philosophical difficulties:

- the problem of becoming,
- the asymmetry of time,
- conscious experience of the present,
- novelty and irreversibility,
- and the status of unrealized possibilities.

These tensions become especially significant once consciousness and realizability are introduced into the ontological picture.

3. TNA and the Problem of Total Closure

The Theory of Axiomatic Necessity distinguishes between two structural domains:

N_0 — Local Operational Dynamics

N_0 corresponds to:

- field equations,
- causality,
- entropy propagation,
- quantum evolution,
- geometry,
- optimization,
- and local physical interactions.

N_0 governs propagation.

N_1 — Realizability Structure

N_1 corresponds to:

- admissibility constraints,
- closure conditions,
- stabilization operators,
- trajectory selection,
- and non-derivable realizability conditions.

N_1 governs realizability itself.

TNA proposes that no sufficiently complex system can generate complete closure solely from N_0 dynamics.

This limitation is formalized as:

Failure of Local Closure

When multiplicity exceeds realizability stabilization capacity:

$$\Delta > 1$$

the system becomes structurally unstable unless constrained by external or non-derivable selection mechanisms.

Under this framework, eternalism represents an attempt at maximal closure entirely within No.

4. Why the Block Universe Becomes Structurally Problematic

The Block Universe successfully describes stabilized relativistic geometry.

However, problems emerge when geometry is treated as ontologically exhaustive.

Several unresolved difficulties arise:

4.1 The Problem of Becoming

Human experience is fundamentally asymmetric and processual.

We do not merely observe ordered spacetime coordinates.

We experience:

- anticipation,
- memory,
- novelty,
- irreversibility,
- and becoming.

Eternalism typically treats this as illusion.

But an illusion that is:

- universal,
- functional,
- coherent,

- and structurally necessary

requires explanation.

4.2 The Problem of Selection

The Block Universe contains all events equally.

But conscious systems experience only realized trajectories.

The ontology therefore lacks a mechanism explaining:

- why certain trajectories become experienced,
- why others remain unrealized,
- and how admissible multiplicity collapses into concrete history.

4.3 The Problem of Asymmetry

Relativistic geometry is largely time-symmetric.

Yet:

- entropy,
- memory,
- causality,
- biological evolution,
- and conscious temporality

all display directional asymmetry.

This asymmetry appears difficult to derive from geometry alone.

5. Smooth Manifolds and the Limits of Closure

Standard eternalism assumes that spacetime forms a globally smooth differentiable manifold:

$$M \in C^\infty$$

Under this assumption:

- propagation is continuous,
- dynamics remain differentiable,
- and all trajectories coexist geometrically.

TNA proposes that this smoothness assumption is insufficient for realizability itself.

Propagation and realizability are not equivalent.

A system may propagate smoothly while remaining incapable of selecting realizable trajectories internally.

TNA therefore proposes that realizability requires non-smooth structure:

$$M \in C^0 \setminus C^1$$

This means:

- continuity is preserved,
- but first derivatives become discontinuous at selection points.

Smooth geometry describes propagation.

Non-smooth structure enables selection.

The Block Universe therefore correctly models stabilized relativistic propagation at scales $\tau \gg T_{Planck}$. Its limitation is not failure but scope: it describes the emergent result of realizability selection, not the process of realizability itself.

6. Introducing Model 2: Beyond the Static Block Universe

Model 2 rejects three extremes:

- fully deterministic eternalism,
- purely stochastic cosmology,
- and completely self-generated smooth spacetime.

Instead, Model 2 proposes:

Reality is a continuously reselected structure whose realizability depends on non-derivable selection events. These events are physically manifest at Planck scale, but structurally present at all scales.

The classical Block Universe emerges only as a macroscopic approximation obtained when underlying selection processes become observationally coarse-grained.

Under this framework:

- relativity remains locally valid,
- local dynamics remain smooth,
- standard physics remains operational between events,
- but **global ontological closure is impossible at every scale, not merely microscopic.**

Becoming therefore regains ontological status **not as a quantum-gravitational correction, but as the fundamental mode of realizability.**

7. Fundamental Ontological Commitment

Under Model 2, reality is not fundamentally a completed four-dimensional manifold.

Instead, spacetime is represented as a continuously updating realizability structure:

$$R(t) = (g_{\mu\nu}(t), \Sigma(t), \Delta(t))$$

where the metric configuration, admissible trajectory set, and structural multiplicity evolve through selection events distributed across all scales.

The apparent Block Universe emerges only after coarse-graining over intervals $\tau \gg T_{Planck}$:

$$M_{block}^4 = \lim_{\tau \gg T_{Planck}} \langle R(t) \rangle_\tau$$

But **the coarse-graining does not eliminate the underlying non-closure; it merely renders it statistically invisible.**

The Planck scale is not the *locus* of non-closure. It is the *threshold* where non-closure becomes mathematically inescapable in physical description.

8. The $\mathcal{N}_1$ Selection Operator

Introduce the realizability selection operator:

$$\hat{S}_n$$

acting on admissible trajectory spaces.

Let:

$$\mathcal{A}_{loc}(\mathcal{O})$$

be the local C^* -algebra of observables associated with bounded open regions $\mathcal{O} \subset M$ of spacetime, in the sense of Haag-Kastler algebraic quantum field theory.

Then:

$$\hat{S}_n \notin \overline{\mathcal{A}_{loc}(\mathcal{O})}^{weak}$$

meaning:

the selection operator cannot be derived from purely local dynamics.

Its properties include:

Non-unitarity

$$\hat{S}^\dagger \hat{S} \neq I$$

Irreversibility

selection introduces asymmetry.

Non-smoothness

selection points generate discontinuities in first derivatives.

Thus, realizability cannot be fully reduced to smooth propagation.

9. Planck-Scale Geometry and C^0 /non- C^1 Structure

Selection occurs at all scales where realizability requires reduction of multiplicity.

The support manifold satisfies:

$$M \in C^0 \setminus C^1$$

at selection points $\{p_n\}$. The Planck-scale density:

$$\rho_{sel} = \frac{1}{T_{Planck}} \approx 1.85 \times 10^{43} \text{ Hz}$$

represents **the minimal resolution at which physical theory can describe selection events without infinite regression**, not the frequency of a fundamental process.

Between described selection events:

$$M|_{(p_n, p_{n+1})} \in C^\infty$$

Thus:

- locally, spacetime remains relativistic,
- globally, realizability requires embedded non-smooth structure **at every scale where selection occurs**.

The Planck frequency is a **lower bound on describability**, not an **upper bound on ontology**.

Instead of idealized delta impulses, realizability transitions occur through compact-support windows:

$$w(t - nT_{Planck})$$

where $w : \mathbb{R} \rightarrow \mathbb{R}^+$ is a smooth compactly supported function with $\text{supp}(w) \subset [-T_{Planck}/2, T_{Planck}/2]$ and $\int w(t)dt = 1$, whose support approximates Planck-scale duration.

10. Temporal Asymmetry as Selection Asymmetry

Under Model 2, temporal asymmetry is not merely thermodynamic.

It emerges structurally from directional realizability selection.

Selection operators satisfy:

$$\hat{S}_n \neq \hat{S}_{n-1}$$

The past corresponds to stabilized realized trajectories.

The future corresponds to admissible multiplicity awaiting selection.

Typically, multiplicity evolves according to:

$$\frac{d\Delta}{dt} \geq 0$$

between selections, as local dynamics generate admissible trajectories faster than they collapse them. Exceptions may occur in strongly dissipative regimes where environmental decoherence reduces multiplicity without N_1 selection. Each selection event locally reduces admissible multiplicity:

$$\Delta^{(n+1)} = 1 < \Delta^{(n)}$$

Entropy growth therefore reflects expanding admissibility, while realizability selection continuously constrains instability.

11. Consciousness as a Resolution-Limited Interface

Consciousness is not itself an N_1 operator.

Rather, consciousness is a macroscopic system whose phenomenology reflects coarse-grained sensitivity to realizability asymmetry.

Human temporal resolution operates at multiple scales:

- Perceptual sensorial resolution: $\tau_{perceptual} \sim 10^{-2}$ s (100 Hz)
- Cognitive conscious resolution: $\tau_{cognitive} \sim 10^{-1}$ s (decision, intention)
- Neural dynamics resolution: $\tau_{neural} \sim 10^{-3}$ s (1 kHz)

Each scale vastly exceeds:

$$T_{Planck} \sim 10^{-44} s$$

Thus consciousness integrates approximately:

$$10^{41} - 10^{44}$$

selection events per second into the appearance of continuous becoming, depending on the temporal scale of integration.

The phenomenological “now” therefore emerges as:

$$Now \equiv \lim_{\tau_{obs} \rightarrow \infty} \bigcup \hat{S}_n(\Sigma^{(n)})$$

The present is not a geometrical slice of static spacetime, but the macroscopic appearance of sub-perceptual realizability updates.

12. Gödel, Closed Timelike Curves, and Admissibility

Gödel demonstrated that Einstein’s equations permit solutions containing closed timelike curves (CTCs).

Under pure eternalism, these trajectories become ontologically problematic because:

- causal order collapses,
- temporal distinction becomes ambiguous,
- and paradoxes emerge.

TNA offers a different interpretation.

CTCs may belong to admissible multiplicity spaces:

$$CTC \in \Sigma^{(n)}$$

while failing realizability selection:

$$CTC \notin \hat{S}_n(\Sigma^{(n)})$$

Thus:

- relativistic geometry remains intact,
- Einstein’s equations remain locally valid,
- but realizability selection prevents paradoxical stabilization.

This preserves physics while avoiding ontological inconsistency.

Under eternalism, CTCs create paradox because cause and effect become ontologically simultaneous. Under Model 2, this paradox dissolves: CTCs are geometrically admissible but structurally unrealizable. The geometry permits; realizability forbids. Einstein's equations describe what can exist geometrically; N_1 selection determines what becomes realized.

13. Why Eternalism Becomes an Emergent Approximation

The Block Universe is not false as geometry. It is **ontologically misleading as cosmology**.

Relativistic spacetime correctly describes causal propagation, local invariance, geodesic structure, and stabilized trajectories.

However, it does not describe — and **cannot** describe, by its own mathematical closure — realizability generation, selection, asymmetry, or ontological becoming.

This limitation is **not a missing quantum-gravity term**. It is a structural feature of any system that attempts self-closure through geometry alone.

The Block Universe is to realizability what thermodynamic equilibrium is to molecular collisions: a valid macroscopic approximation that becomes misleading when mistaken for fundamental ontology.

14. Conclusion

This paper has proposed a reinterpretation of eternalism through the Theory of Axiomatic Necessity.

Under Model 2:

- spacetime remains locally relativistic,
- geometry remains operationally valid,
- standard physics remains preserved between selection events.

But **reality is not fundamentally a completed static manifold at any scale**.

The apparent Block Universe emerges from continuous realizability selection whose non-closure is **structurally present at all scales** and **physically manifest at Planck scale**.

Temporal becoming regains ontological status not as a quantum correction, but as the fundamental asymmetry of realizability itself.

Reality is not a pre-existing completed block. It is a continuously stabilized structure whose apparent geometric completeness emerges from non-derivable realizability selection — present everywhere, visible at the threshold.

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